# HDOC Day-45

## Name- Aaryan pratap singh

## Sap id - 590032842

## Algorithm

### Main Program

1. Start.
2. Display the menu:
   - Change date format from `dd/mm/yyyy` to `dd-mmm-yy`.
   - Print all substrings of a string.
   - Exit.
3. Read the user's choice.
4. Use `switch-case` to call the appropriate function.
5. Display an error message for an invalid choice.
6. Repeat until the user selects Exit.
7. Stop.

### Operation 1 — Change Date Format

1. Read the date in `dd/mm/yyyy` format.
2. Extract the day, month, and year using `sscanf()`.
3. Check whether the month is between 1 and 12.
4. Store month abbreviations in a two-dimensional character array.
5. Use the month number to obtain the corresponding month abbreviation.
6. Take the last two digits of the year using `year % 100`.
7. Display the date in `dd-mmm-yy` format.

Example:

```
Input : 12/10/2026
Output: 12-Oct-26
```

### Operation 2 — Print All Substrings

1. Read a string.
2. Find its length.
3. Select each character as the starting position of a substring.
4. For every starting position, extend the substring one character at a time.
5. Print the characters between the starting and ending positions.
6. Continue until all possible substrings are displayed.

Example:

```
Input : abc

Output:
```

```
a
ab
abc
b
bc
c
```

## Test-Case Table

| Test | Choice | Input | Expected Output |
|---|---|---|---|
| 1 | 1 | `12/10/2026` | `12-Oct-26` |
| 2 | 1 | `01/01/2025` | `01-Jan-25` |
| 3 | 1 | `25/12/2026` | `25-Dec-26` |
| 4 | 1 | `15/06/2030` | `15-Jun-30` |
| 5 | 1 | `12/13/2026` | Invalid date |
| 6 | 2 | `abc` | `a, ab, abc, b, bc, c` |
| 7 | 2 | `ab` | `a, ab, b` |
| 8 | 2 | `xyz` | `x, xy, xyz, y, yz, z` |
| 9 | 3 | — | Program exits |
| 10 | 5 | — | Invalid menu choice |

## C Program

```
#include <stdio.h>

#include <string.h>

#define MAX 100

void changeDateFormat(void);

void printSubstrings(void);

int main(void)
```

```
{
    int choice;
    do
    {
        printf("\n===== HDOC Day-45 =====\n");
        printf("1. Change date format\n");
        printf("2. Print all substrings of a string\n");
        printf("3. Exit\n");
      printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();
      switch (choice)
        {
          case 1:
                changeDateFormat();
                break;
         case 2:
                printSubstrings();
                break;
            case 3:
                printf("Exiting program...\n");
                break;
```

```
            default:
                printf("Invalid choice. Please enter 1, 2, or 3.\n");
        }
    } while (choice != 3);
    return 0;
}
/* Function to change date format */
void changeDateFormat(void)
{
    char date[20];
    int day, month, year;
    char months[12][4] = {
        "Jan", "Feb", "Mar", "Apr",
        "May", "Jun", "Jul", "Aug",
        "Sep", "Oct", "Nov", "Dec"
    };
    printf("Enter date in dd/mm/yyyy format: ");
    fgets(date, sizeof(date), stdin);
    if (sscanf(date, "%d/%d/%d",
               &day, &month, &year) != 3)
    {
        printf("Invalid date format.\n");
```

```
        return;
    }
   if (month < 1 || month > 12)
    {
        printf("Invalid date.\n");
        return;
    }
    printf("Formatted date: %02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}
/* Function to print all substrings */
void printSubstrings(void)
{
    char str[MAX];
    int i, j, k;
    int length;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';
    length = strlen(str);
    printf("All substrings are:\n");
    for (i = 0; i < length; i++)
```

```
    {
        for (j = i; j < length; j++)
        {
            for (k = i; k <= j; k++)
            {
                printf("%c", str[k]);
            }
            printf("\n");
        }
    }
}
```

# EXECUTION

